Assessment: Build a Multi-Agent AI News Assistant Using LangChain

Business Scenario

An organization wants an AI assistant that researches the latest developments in a technology topic, summarizes the findings, and generates a notification for employees.

Objective

Build a multi-agent system using **LangChain and LangGraph** with three agents.

Agent 1 – Research Agent

- Accept a topic from the user (e.g., "Latest developments in Generative AI").
- Search and collect relevant information from the web.
- Pass the research findings to the next agent.

Agent 2 – Summarizer Agent

- Receive the research findings.
- Generate a concise summary containing 5 key points.
- Include relevant source links.

Agent 3 – Notifier Agent

- Receive the summarized information.
- Generate a professional email notification.
- Save the notification to a text file.

Agile Team Assignment

Before implementation:

1. Create one Epic for the overall project.
2. Break the Epic into Features and User Stories.
3. Divide the work into Tasks.
4. Assign tasks to different team members.
5. Define acceptance criteria for each user story.

Technical Requirements

- Use Python, LangChain, and LangGraph.
- Implement three separate agents.
- Coordinate agent execution using LangGraph.
- Pass information between agents.
- Handle basic errors and missing information.
- Demonstrate the complete workflow.

Expected Deliverables

1. Agile backlog with assigned responsibilities.
2. Architecture diagram showing agent communication.
3. Working Python application.
4. Sample execution demonstrating all three agents.
5. README explaining setup and execution.

Final Outcome: A user enters a topic, and the system automatically researches it, summarizes the findings, and prepares an email notification.